

ADDITION OF SWORDFISH DISTRIBUTION MODEL TO LONGLINE SIMULATOR STUDY

Francesca C. Forrestal^{1*} and Michael J. Schirripa²

SUMMARY

A species distribution model (SDM) for swordfish was developed through habitat suitability indices and pop-up satellite tag data. The model used detailed biological and oceanographic data to define the spatial distribution of swordfish. This SDM was compared to a similar model developed previously for blue marlin. The two species SDMs were used as input into the longline simulator (LLSIM) to obtain catch per unit effort datasets based on the US pelagic longline fishery. These datasets were used to identify methods to reduce blue marlin bycatch through the use of time-area closures. The swordfish catch and effort data were also used to assess different methods of CPUE standardization. The datasets produced by LLSIM have a known, underlying population abundance and can be used to test the accuracy of standardization methods that are commonly used in assessments of swordfish. Model approaches used in this report include a Tweedie generalized linear model, a negative binomial generalized linear model, a lognormal generalized linear model and Delta lognormal generalized linear model. Standardized indices obtained from these models were compared to the known population abundance entered into LLSIM.

RÉSUMÉ

Un modèle de distribution des espèces (SDM) pour l'espadon a été mis au point à partir d'indices d'adéquation de l'habitat et de données de marques satellites de type pop-up. Le modèle a utilisé des données biologiques et océanographiques détaillées pour définir la distribution spatiale de l'espadon. Ce SDM a été comparé à un modèle similaire développé précédemment pour le makaire bleu. Les SDM pour les deux espèces ont été utilisés comme donnée d'entrée dans le simulateur palangrier (LLSIM) pour obtenir des jeux de données de prise par unité d'effort reposant sur la pêche palangrière pélagique américaine. Ces jeux de données ont été utilisés pour identifier des méthodes permettant de réduire les prises accessoires de makaire bleu par le biais de fermetures spatio-temporelles. Les données de prise et d'effort de l'espadon ont également été utilisées pour évaluer différentes méthodes de standardisation de la CPUE. Les jeux de données produits par la LLSIM ont une abondance de population sous-jacente connue et peuvent être utilisés pour tester l'exactitude des méthodes de standardisation qui sont couramment utilisées dans les évaluations de l'espadon. Les approches de modèle utilisées dans ce rapport comprennent un modèle linéaire généralisé Tweedie, un modèle linéaire généralisé binomial négatif, un modèle linéaire généralisé lognormal et un modèle linéaire généralisé lognormal Delta. Les indices standardisés obtenus à partir de ces modèles ont été comparés à l'abondance connue de la population saisie dans le LLSIM.

RESUMEN

Se desarrolló un modelo de distribución de especies (SDM) para el pez espada mediante índices de idoneidad del hábitat y datos de marcas pop-up por satélite. El modelo usaba datos biológicos y oceanográficos detallados para definir la distribución espacial del pez espada. Este SDM se comparó a un modelo similar desarrollado previamente para la aguja azul. Los SDM de dos especies se utilizaron como entrada en el simulador de palangre (LLSIM) para obtener conjuntos de datos de captura por unidad de esfuerzo basados en la pesquería de palangre pelágico de Estados Unidos. Los conjuntos de datos se utilizaron para identificar métodos para reducir la captura fortuita de aguja azul mediante el uso de vedas espacio-temporales. Los datos de captura

¹ CIMAS, Rosenstiel School of Marine and Atmospheric Science, 4600 Rickenbacker Cwy., Miami, FL 33149. fforrestal@miami.edu

² NOAA Fisheries, Southeast Fisheries Center, Sustainable Fisheries Division, 75 Virginia Key Dr., Miami, FL 33149.

Michael.Schirripa@noaa.gov

* Corresponding author

y esfuerzo del pez espada se utilizaron también para evaluar diferentes métodos de estandarización de la CPUE. Los conjuntos de datos producidos por el LLSIM tienen una abundancia de población subyacente conocida y pueden usarse para probar la precisión de los métodos de estandarización que se usan habitualmente en las evaluaciones de pez espada. Los enfoques de modelos usados en este informe incluyen un modelo lineal generalizado Tweedie, un modelo lineal generalizado binomial negativo, un modelo lineal generalizado lognormal y un modelo lineal generalizado Delta lognormal. Los índices estandarizados obtenidos a partir de estos modelos se compararon con la abundancia de población conocida introducida en el LLSIM.

KEYWORDS

Swordfish, CPUE Standardization, Bycatch, Blue marlin, Species Distribution, Statistics

1. Introduction

Longline fisheries operate throughout the world's oceans, targeting and catching a wide variety of species. Fleets target species by varying gear characteristics and spatio-temporal considerations. Despite this, many species can be incidentally caught by longline gear as it is a non-selective gear and a variety of species can encounter the hooks during various time periods of fishing operations (Bigelow *et al.*, 2006; Goodyear, 2003). The effort, locations and gear configurations have also changed over time, due to shifts in target species distributions, changing management regulations and technological improvements (Torres-Irribe *et al.*, 2014). These changes in effort can generate issues in using catch and effort datasets that are applied by fishery scientists to assess the status of fish stocks. Catch per unit effort (CPUE) data are routinely used in stock assessments as the assumption that catch is related to effort, the catchability of the gear and the abundance or number of fish. However, any variation in the effort or location can bias the CPUE trend over time, leading to the necessity of standardizing the CPUE to reduce the effect of these shifts as well as minimize bias (Maunder *et al.*, 2006; Ortiz and Arocha, 2004).

Choosing the best methods to standardize CPUE datasets is challenging as the underlying population is generally unknown. The development of species distribution models that can be paired with simulated fishing effort allows for detailed explorations of different CPUE approaches. This paper details the developments of a swordfish (*Xiphias gladius*) distribution model (SDM) that can be used as an input to the longline simulator previously developed by C. Philip Goodyear (Forrestal *et al.*, 2019; Goodyear, 2016b). A SDM had been previously developed and the addition of swordfish, a target in the longline fishery, also allows for examinations of target and bycatch species interactions using LLSIM (Goodyear, 2016a).

Blue marlin (*Makaira nigricans*) are incidentally caught as bycatch in the Atlantic by the pelagic longline fishery. The recent assessment conducted by ICCAT classified blue marlin as both overfished and experiencing overfishing (Anon, 2018). Efforts aimed at diminishing the incidental take of blue marlin by the longline fishery have centered on time-area closures and adaptations to gear configuration while at the same time limiting reductions to target catches, such as swordfish. Real world testing of different bycatch reduction approaches can be costly and logistically challenging. Pairing the blue marlin and swordfish SDMs with the LLSIM allows for methods of bycatch reduction to be explored.

2. Materials and methods

2.1 Addition of SWO to SDM

We built a swordfish Species Distribution Model (SDM) based on the methods outlined in Goodyear (2015). The SDM describes the spatial and temporal distribution of habitat using species specific affinity curves for the habitat variables utilized. The SDM was dimensioned by 1x1 degree latitude and longitude, 46 depth layers, year and month. Oceanographic variables were obtained from the Community Earth System Model (CESM) (Danabasoglu *et al.* 2012). These five measurements were used to create individual three-dimensional habitat "cubes" for the years and months 1986-2005. The SDM requires detailed information on habitat use by swordfish and this information was obtained from pop-up satellite (PSAT) data from swordfish in the Atlantic. In cooperation with other ICCAT CPCs (Portugal, Canada, Spain and USA) PSAT data was contributed and shared via a single OwnCloud website, this effort is intended to be continued as more PSAT data becomes available. Habitat variables were chosen based on those fulfilling the broad environmental and biological needs of swordfish including depth, oxygen, temperature and forage. Habitat variables specific affinity curves were built from either observational data or published values. These affinity curves were subsequently used to estimate the swordfish habitat suitability and abundance of each modeled cube.

Ocean depth. The depth of the ocean is considered a habitat variable. Presently only latitude-longitude cells that average 200 m or greater are included and no additional weighting for ocean depth is considered in determining the areal distribution. However, the observed proportions from swordfish monitored with satellite tags partition the vertical distribution of swordfish within the volume of water at a location day and night (**Figure 1A**).

Oxygen: Although brief excursions into hypoxic conditions are possible or even routine, at some level the concentration of oxygen must be a limiting factor. A precise model of oxygen suitability is not available for swordfish, but observations from several studies indicate swordfish are more tolerant of low DO than many other pelagic species (Abascal et al., 2010; Abecassis et al., 2012; Braun et al., 2015; Dewar et al., 2011). Pending a more accurate analysis, the oxygen selectivity curve used for the blue marlin SDM in Goodyear (2016) was modified for swordfish by adjusting the transition downward by 1 ml/l so that the median corresponds to 2.5 ml/l (**Figure 1B**).

Temperature: Swordfish exhibit a wide thermal plasticity in occupied habitat. They also show a well-documented propensity for spending daytime hours deeper in the water column (Abascal et al., 2010; Abecassis et al., 2012; Braun et al., 2015; Dewar et al., 2011; Neilson et al., 2009; Neilson et al., 2014). This trend was also evident in the tag data available for this study and results in a strong diel cycle in the temperatures of the habitat utilized (**Figure 1C**). The large amount of time spent in cooler environments during the daytime subverts using the temperature utilization pattern to map the spatial extent of suitable habitat from temperature as was done for blue marlin. This is partly because the lower daytime mode of temperature occurs at temperatures present in nearly every latitude-longitude ocean grid, and partly because like temperatures extend beyond the modeled region. Nonetheless, temperature is an important determinant of suitable habitat.

It is reasonably clear that the temperatures at which swordfish spend daytime hours in **Figure 1C** are not limiting areal distributions because such temperatures are nearly everywhere present in the water column. In contrast, the nighttime temperature utilization pattern may have predictive value, and we apply the relative proportions at temperatures from 15° C and above as weightings for the temperature selectivity curve for use in the SDM to obtain the 2-D spatial distribution pattern. The method used here to predict the day/night distribution from the global observed depth distributions pattern can predict swordfish to be in depths that are either too warm or cold based on the local oceanographic conditions at time and place. The current model avoids that possibility with a second HSI model of temperature tolerances imposed on the vertical predictions to limit the values to the range of observed values (**Figure 1D**).

Zooplankton. Prey items that swordfish directly target would have been the ideal habitat variable to include in the model. However, distribution of such prey (squid, bait fish, etc.) were not available for the spatial and temporal dimensions of the SDM. Instead, concentration of zooplankton carbon, summed through the modeled water column, was used as a proxy for swordfish prey items present further up the food chain. To more accurately reflect the time it takes zooplankton carbon to work its way through the food chain and better represent swordfish prey, zooplankton carbon was staggered by four months. This resulted in an accurate depiction of the known migration of swordfish from southern to more northern latitudes during the summer and fall months (months of increased feeding) (Schirripa et al. 2016).

The zooplankton affinity curve was initially developed by using a logistic asymptotic curve and assuming that higher densities of zooplankton were more preferred than lower densities (**Figure 1E**). The inflection point of curve was arrived upon by comparing the estimated density of habitat to the observed density of catch throughout the range of the model using a least squares approach.

Sea surface gradient. Swordfish tend to congregate around temperature fronts. Podesta et al. (1993) found an association between swordfish catch-per-unit effort (CPUE) and distance to the nearest front. These fronts develop from warm water meeting colder water and can be located by the presence of relatively large changes in sea surface height gradients. We used the square root of the maximum difference in sea surface height between a grid and its four adjacent grids to calculate a sea surface gradient (SSG).

The affinity curve for SSG was developed by calculating the percentage of time that each PSAT tagged fish spent in each of the SSG bins and fitting a polynomial function to the resulting distribution (**Figure 1E**).

2.2 Model Validation and spatial distribution overlap with blue marlin

The SDM of swordfish was validated through the use of ICCAT Task II swordfish catch data as well as observer data and PSAT tracks. Further validation was achieved by correlating average swordfish catch distribution from the Task II database to the average SDM distribution. Maps were produced using R and the packages *corrplot*, *ggplot2*, *gstat*, *rgdal*, *maptools* and *sp* (Pebesma, 2004; Pebesma and Bivand, 2005; Bivand *et al.*, 2013; Gräler *et al.*, 2016; Wei and Simko, 2016; Wickham, 2016; Bivand *et al.*, 2017; Bivand and Lewin-Koh, 2017).

2.3 LLSIM catch datasets

The longline simulator was ran using two species distributions derived from the SDM, one for blue marlin and the other for swordfish. The underlying population abundance in numbers for both species was obtained from the most recent Stock Synthesis assessment models for blue marlin and swordfish (Anonymous, 2018, 2017). The catches produced by LLSIM were compared to the logbook catches in number for the two species (**Figure 2** and **Figure 3**). This was to ensure that the LLSIM catch and logbook catches were within the same order of magnitude to provide a realistic and useful dataset (Goodyear, 2016b).

Two effort datasets were produced, one with a “status quo” level of fishing effort and one with a time-area closure. The time-area closure dataset was developed to examine methods of bycatch reduction. Areas with high levels of CPUE for blue marlin were identified as possible locations for a closure. For this exercise, a spatial-temporal closure was chosen in the region of Western Caribbean and Florida Straits (Fig.). During the closure, all longline effort was removed in LLSIM. The “status quo” dataset contained the fishing effort that mimicked the US pelagic longline fishing effort from 1986-2012 (Forrestal *et al.*, 2019). The effort in the time-area closure dataset remained the same overall in terms of set numbers, baskets, and gear configuration as the “status-quo” dataset. However, the effort in the closed area location was shifted to outside of the closed area for the 4th quarter: October, November and December (**Figure 4** and **Figure 5**).

2.4 CPUE standardization

The CPUE standardizations were conducted on the full datasets produced by LLSIM, which comprised 297,180 sets from 1986-2012. The sets from 1986 only came from the final quarter of 1986 (October, November and December). Information for each set included the year, latitude, longitude, month, year, number of light sticks, hook type, bait type, hooks between floats (hbf), total hooks, and catches of swordfish and blue marlin in numbers (**Table 1**). Latitude and longitude were grouped by area (**Figure 5**) and months were grouped into quarters. Light sticks numbers were previously binned in the LLSIM model construction and included 0 light sticks, 1-500, 501-1500 and unknown. Hook types included j-hooks, circle hooks, a combination of j and circle hooks and unknown hook types. Bait type included live, dead, artificial and unknown. CPUE was calculated for swordfish as the number caught/1000 hooks. The standardizations used included four different generalized linear models (GLMs) and used CPUE as the response variable.

Swordfish are a target of the US pelagic longline fleet and as such, there are few sets that do not contain swordfish catches (**Table 2**). The LLSIM dataset produced an average of 86% sets that contained swordfish but the amount of zero catches can cause issues with standardization as they do not fit standard distributions (**Figure 6**). Three of the four approaches are more flexible for zero-inflated data, the Tweedie GLM, negative binomial GLM and the delta GLM. The fourth approach used was a lognormal GLM. These four approaches are all commonly used for standardizing swordfish CPUE used in ICCAT assessments (Coelho *et al.*, 2017; Laretta and Walter, 2017; Ramos-Cartelle *et al.*, 2017; Su and Sun, 2017).

The Tweedie distribution belongs to the exponential family of distributions and has shown to handle zero-catch data in a cohesive manner (Shono, 2008). The Tweedie GLM utilized a gamma variance power and the logarithmic link function and was run using the Tweedie package in R (Dunn, 2014). The negative binomial GLM assumes a negative binomial distribution. The lognormal GLM is not specific to zero-inflated data but is routinely used in CPUE standardizations (Hinton and Maunder, 2003). The Gaussian distribution was used with a log link. A constant was added to the CPUE response variable to account for the presences of zeroes in the data. The constant was calculated as 10% of the mean of the catch rate (Campbell, 2004). The Delta GLM employs two models that are combined to retrieve the yearly index. The data is split into presence absence data where sets with positive catches for swordfish are counted as 1 and sets with no swordfish are zeroes. This data was modeled using a GLM with a binomial error structure and logit link function. Data with positive catches is then modeled with a GLM Gamma distribution and log link. The negative binomial GLM, lognormal GLM and delta GLMs were run using the R package MASS (Venables and Ripley, 2002).

Models:

$$\text{CPUE} \sim \text{factor(Year)} + \text{factor(Quarter)} + \text{Area} + \text{factor(Hook type)} + \text{factor(Bait type)} + \text{factor(hbf)} + \text{factor(light)}$$

Another set of models were run incorporating the corresponding sea surface temperature (SST) for each set. This data was obtained from the CESM oceanographic data.

$$\text{CPUE} \sim \text{factor(Year)} + \text{factor(Quarter)} + \text{Area} + \text{factor(Hook type)} + \text{factor(Bait type)} + \text{factor(hbf)} + \text{factor(light)} + \text{SST}$$

The indices of abundance were estimated as the product of the least squares means using the lsmeans R package (Lenth, 2016) or the estimated marginal means with the emmeans package as were estimates of uncertainty (Lenth, 2016, 2019). Indices of abundance for each model were scaled to the standardized mean for comparison. The fit of the models to the true population abundance of swordfish entered were LLSIM were calculated using the root mean square error (RMSE). This value was compared to the mean coefficient of variation for each model.

3. Results

3.1 Addition of SWO to SDM

Model validation. We qualitatively validated the SDM estimated annual mean spatial density of swordfish habitat by overlying the available PSAT tracks and direct onboard observations from several CPC's observer programs. There was a very high visual correlation between the estimated habitat densities and the PSAT and onboard observations (**Figure 6**). It should be noted that neither PSTs nor observer coverage are designed to have representative coverage of the Atlantic Ocean. As such, there can be areas estimated to be suitable swordfish habitat yet do not have either PSAT tracks or onboard observer observations. Consequently, PSATs and onboard observations overlapping area of high estimated swordfish densities are indicative of an accurate model fit, but areas of high estimated swordfish densities without overlying PSATs or onboard observations are not indicative of a lack of an accurate fit.

3.2 Spatial distribution overlap with blue marlin and bycatch reduction

Blue marlin and swordfish had the greatest overlap in the tropical Western Atlantic (**Figure 7** and **Figure 8**). This is reflected in both the logbook and LLSIM datasets as the largest blue marlin CPUE occurs in that spatial-temporal zone. The use of a closed area in that region for 2005-2008 resulted in a reduction in blue marlin catches and an increase in swordfish catches (**Table 3**).

3.3 CPUE standardization

Across the eight models, all the explanatory variables included were significant for standardizing the swordfish CPUE produced by LLSIM (**Table 4** and **Table 5**). Factors and variables that contributed the most deviance varied across the model approaches but both year and area were large contributors for all approaches examined. Standardized CPUE indices for all approaches are shown in **Tables 6-9**.

The standardized CPUEs from the eight models examined exhibited similar patterns across the time series with a slightly decreasing trend from 2010 (**Figure 9**). The exception to this was the Tweedie model that had a slightly flattened overall time series as compared to the other models. The trends between the same model approaches with SST and without were similar though there were slight differences. Overall, the Tweedie GLM had the lowest average CV% (0.662 with SST and 0.665) while the negative binomial with SST and the Delta GLM with SST had the highest CV% (3.281 and 3.061) (**Table 10**). The negative binomial and Delta GLM models that contained SST also had a higher CV% than the negative binomial GLM and Delta GLM models that did not contain SST while the Tweedie and Lognormal GLMs that did not contain SST had a slightly lower average CV% than the models that did contain SST.

The standardized CPUE with the best fit to actual population abundance according to the RMSE was the negative binomial GLM (0.047 for both models) while the Tweedie GLM had the worst fit (0.100 + SST and 0.101) (**Table 10** and **Figure 10**). This is the exact opposite result seen in the average CV% for each model. Examining the diagnostic plots for the Tweedie GLM and negative binomial GLM demonstrated that the negative binomial GLM had a better fit to the residuals than that of the Tweedie (**Figures 12-14**).

4. Discussion

The SDM developed for this study showed very good agreement between the observed spatial distribution and the predicted spatial distribution of swordfish habitat. Furthermore, the within year migratory patterns of swordfish were also reproduced with good agreement. We therefore conclude that the SDM is acceptable to use as a base distribution model as input into the longline simulator (LLSIM). The use of both a target species and bycatch species in LLSIM will allow various explorations to be made around CPUE standardizations, bycatch minimization, and whether information about oceanographic conditions is useful for avoiding bycatch.

Unlike blue marlin, swordfish have a remarkable plasticity in their use of available habitat (Goodyear 2016, Braun et al., 2015). Swordfish are more tolerant of both lower temperatures and lower oxygen levels and inhabit deeper depths than do blue marlin. This led us to the conclusion that blue marlin habitat would be the limiting factor in habitat overlap. Examination of the blue marlin habitat mapping shows the highest densities of blue marlin habitat exists between approximately latitudes 20 degrees south and 20 degrees north, predominantly in the western Atlantic (Goodyear 2016). Examination of the swordfish habitat mapping shows that within this area, the highest density of swordfish habitat is found between approximately latitude 15 degrees south and 20 degrees north. Thus, the area of greatest habitat overlap would be in the Caribbean Sea area and the areas off the eastern coast Florida, USA.

The use of the spatial-temporal closure did achieve the aim of reducing the amount of blue marlin caught. While the reduction was minimal in terms of fewer blue marlin caught, the effort in this exercise was from a single fleet. It is reasonable to expect the reduction could be greater when other fishing fleets are taken in account. The time-area closure had a positive effect on the amount of swordfish caught as the target species. One of the criticisms of time-area closures is the potential to reduce the catch of target species which did not occur in this exercise (O'Keefe et al., 2014). This is only an initial exploration of identifying possible means of reducing blue marlin bycatch and more work is needed in this area.

The use of LLSIM datasets allows for the unique opportunity to assess CPUE standardization model performance in terms of both traditional model diagnostics and fit to the known abundance trend. Comparing the average CV % and the RSME results from the Tweedie GLM and the negative binomial GLM highlight an apparent conflict between model selection. RMSE values are not generally used as model performance diagnostics for CPUE standardizations in this manner as the population abundance is an unknown value but this exercise does offer an interesting perspective. The lowest average CV% for the models was obtained from the Tweedie GLM while the highest was that of the negative binomial GLM, suggesting that the Tweedie was a more precise fit to the catch and effort data. The lowest RMSE values, as a metric of model accuracy, were obtained from the negative binomial GLM while the Tweedie had the largest RMSE. However, precision does not mean the measurement is more accurate and these results capture this well.

Acknowledgements

This work was carried out under the provision of the ICCAT Science Envelope and the partially funded by the European Union through the EU Grant Agreement No. SI2.796428 - Strengthening the scientific basis for decision-making in ICCAT. F. C. Forrestal's contribution was also supported by the NOAA Southeast Fisheries Science Center via the University of Miami Cooperative Institute for Marine and Atmospheric Science. The authors wish to thank Sang-ki Lee and Yanyun Liu of NOAA Atlantic Oceanographic and Meteorological Laboratory for gathering and compiling the oceanographic data. We also wish to thank Guillermo Diaz and Allison Shideler for gathering and compiling the US longline effort data.

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Table 1. Variables used in the CPUE standardizations, variable type: categorical or continuous and the levels or range included. Latitude and longitude in one ° cells, HBF=hooks between floats, SST (°C) = sea surface temperature.

Variable	Type	Range
Year	Categorical	1986-2015
Month	Categorical	1-12
Lat.	Continuous	-30°S-53°N
Long.	Continuous	-95°W-15°E
HBF	Categorical	2-6
Hook	Categorical	1-4
Bait	Categorical	1-4
Lights	Categorical	0-3
SST	Continuous	2-31

Table 2. Summary statistics of swordfish catches generated by LLSIM.

Year	Sets	Prop Pos	SWO (number)	hooks ('000s)	Nominal CPUE
1986	1,352	90%	10,743	533	22.05
1987	10,526	85%	73,483	4,795	17.85
1988	12,270	90%	103,297	5,275	20.29
1989	13,784	92%	136,930	5,776	24.08
1990	12,966	91%	127,576	5,443	24.17
1991	12,643	90%	119,168	6,007	21.37
1992	12,317	88%	113,181	6,539	19.05
1993	12,041	89%	89,367	6,789	14.89
1994	12,885	89%	86,163	7,487	13.10
1995	13,749	85%	79,224	8,713	10.14
1996	13,891	86%	81,940	8,926	10.07
1997	12,938	86%	84,468	8,358	10.77
1998	10,875	87%	83,430	7,103	12.60
1999	10,279	85%	76,528	6,713	12.19
2000	10,108	84%	80,386	6,587	12.74
2001	9,640	86%	98,124	6,812	14.28
2002	8,665	84%	94,117	6,211	14.89
2003	8,458	81%	77,494	6,244	12.14
2004	8,782	79%	64,385	6,486	9.78
2005	7,115	81%	52,632	5,319	9.50
2006	6,957	83%	57,201	5,161	10.66
2007	7,700	78%	51,751	5,409	9.32
2008	8,245	84%	67,767	5,990	11.03
2009	8,423	80%	62,392	6,285	9.69
2010	7,200	88%	61,953	5,459	10.97
2011	7,601	89%	61,318	5,577	10.69
2012	9,810	81%	65,922	7,133	9.06

Table 3. Catches from “status quo” effort LLSIM run as compared to catches of from closed area effort LLSIM run for BUM and SWO. Closed area in effect from 2005-2008 for the fourth quarter.

BUM				SWO		
Year	Status Quo	Closed Area	% Δ	Status Quo	Closed Area	% Δ
2005	641	645	0.6	52632	53042	0.8
2006	486	481	-1.0	57201	58506	2.2
2007	558	550	-1.5	51751	52861	2.1
2008	703	663	-6.0	67767	68783	1.5

Table 4. Deviance tables for models containing SST, including the degrees of freedom (Df), the deviance, the residual degrees of freedom (Resid Df), the residual deviance (Resid Dev) and the degree of significance (Pr(>Chi)).

Model	Df	Deviance	Resid Df	Resid Dev	Pr(>Chi)
Delta GLM - binomial					
Intercept only			271219	219829	
Year	26	2975	271193	216853	< 2.2e-16
Quarter	3	53	271190	216800	1.47E-11
Area	9	40728	271181	176072	< 2.2e-16
Hook type	4	745	271177	175327	< 2.2e-16
Bait type	3	1113	271174	174215	< 2.2e-16
HBF	4	258	271170	173956	< 2.2e-16
Lightsticks	3	11750	271167	162206	< 2.2e-16
SST	1	1272	271166	160934	< 2.2e-16
Delta GLM - lognormal					
Intercept only			233204	30861160	
Year	26	5925014	233178	24936146	< 2.2e-16
Quarter	3	415098	233175	24521048	< 2.2e-16
Area	9	2258429	233166	22262619	< 2.2e-16
Hook type	4	1370283	233162	20892336	< 2.2e-16
Bait type	3	244760	233159	20647576	< 2.2e-16
HBF	4	3285330	233155	17362246	< 2.2e-16
Lightsticks	3	5955372	233152	11406875	< 2.2e-16
SST	1	189555	233151	11217319	< 2.2e-16
Lognormal GLM					
Intercept only			271219	39839428	
Year	26	6301023	271193	33538405	< 2.2e-16
Quarter	3	365650	271190	33172756	< 2.2e-16
Area	9	5790403	271181	27382352	< 2.2e-16
Hook type	4	1444007	271177	25938345	< 2.2e-16
Bait type	3	306350	271174	25631995	< 2.2e-16
HBF	4	3241153	271170	22390842	< 2.2e-16
Lightsticks	3	6914885	271167	15475957	< 2.2e-16
SST	1	29554	271166	15446403	< 2.2e-16
Tweedie GLM					
Intercept only			271219	528477	
Year	26	27987	271193	500490	< 2.2e-16
Quarter	3	908	271190	499582	< 2.2e-16
Area	9	31539	271181	468043	< 2.2e-16
Hook type	4	4137	271177	463906	< 2.2e-16
Bait type	3	4599	271174	459307	< 2.2e-16
HBF	4	17083	271170	442223	< 2.2e-16
Lightsticks	3	99271	271167	342952	< 2.2e-16
SST	1	91	271166	342861	< 2.2e-16
Negative Binomial GLM					
Intercept			271219	792636	
Year	26	75043	271193	717592	< 2.2e-16
Quarter	3	2458	271190	715135	< 2.2e-16
Area	9	80972	271181	634163	< 2.2e-16
Hook type	4	11115	271177	623049	< 2.2e-16
Bait type	3	9986	271174	613063	< 2.2e-16
HBF	4	43239	271170	569824	< 2.2e-16
Lightsticks	3	209903	271167	359921	< 2.2e-16
SST	1	197	271166	359724	< 2.2e-16

Table 5. Deviance tables for models without SST, including the degrees of freedom (Df), the deviance, the residual degrees of freedom (Resid Df), the residual deviance (Resid Dev) and the degree of significance (Pr(>Chi)).

Model	Df	Deviance	Resid Df	Resid Dev	Pr(>Chi)
Delta GLM - binomial					
Intercept only			271219	219829	
Year	26	2975	271193	216853	< 2.2e-16
Quarter	3	53	271190	216800	1.47E-11
Area	9	40728	271181	176072	< 2.2e-16
Hook type	4	745	271177	175327	< 2.2e-16
Bait type	3	1113	271174	174215	< 2.2e-16
HBF	4	258	271170	173956	< 2.2e-16
Lightsticks	3	11750	271167	162206	< 2.2e-16
Delta GLM - lognormal					
Intercept only			233204	30861160	
Year	26	5925014	233178	24936146	< 2.2e-16
Quarter	3	415098	233175	24521048	< 2.2e-16
Area	9	2258429	233166	22262619	< 2.2e-16
Hook type	4	1370283	233162	20892336	< 2.2e-16
Bait type	3	244760	233159	20647576	< 2.2e-16
HBF	4	3285330	233155	17362246	< 2.2e-16
Lightsticks	3	5955372	233152	11406875	< 2.2e-16
Lognormal GLM					
Intercept only			271219	39839428	
Year	26	6301023	271193	33538405	< 2.2e-16
Quarter	3	365650	271190	33172756	< 2.2e-16
Area	9	5790403	271181	27382352	< 2.2e-16
Hook type	4	1444007	271177	25938345	< 2.2e-16
Bait type	3	306350	271174	25631995	< 2.2e-16
HBF	4	3241153	271170	22390842	< 2.2e-16
Lightsticks	3	6914885	271167	15475957	< 2.2e-16
Tweedie GLM					
Intercept only			271219	528477	
Year	26	27987	271193	500490	< 2.2e-16
Quarter	3	908	271190	499582	< 2.2e-16
Area	9	31539	271181	468043	< 2.2e-16
Hook type	4	4137	271177	463906	< 2.2e-16
Bait type	3	4599	271174	459307	< 2.2e-16
HBF	4	17083	271170	442223	< 2.2e-16
Lightsticks	3	99271	271167	342952	< 2.2e-16
Negative Binomial					
Intercept only			271219	792023	
Year	26	74969	271193	717055	< 2.2e-16
Quarter	3	2455	271190	714599	< 2.2e-16
Area	9	80894	271181	633705	< 2.2e-16
Hook type	4	11103	271177	622602	< 2.2e-16
Bait type	3	9977	271174	612624	< 2.2e-16
HBF	4	43198	271170	569426	< 2.2e-16
Lightsticks	3	209733	271167	359693	< 2.2e-16

Table 6. Nominal and standardized CPUE estimates (numbers/hooks) for the tweedie GLM model with sea surface temperature and without. Least squared mean point estimated, CV% and 95% CI listed.

Tweedie GLM + SST					
Year	Nominal	LS mean	CV%	LCL	UCL
1986	22.05	2.33	1.00	2.28	2.37
1987	17.85	2.11	0.70	2.08	2.14
1988	20.29	2.13	0.68	2.10	2.16
1989	24.08	2.28	0.58	2.25	2.31
1990	24.17	2.32	0.58	2.29	2.35
1991	21.37	2.29	0.58	2.27	2.32
1992	19.05	2.27	0.59	2.24	2.29
1993	14.89	2.18	0.62	2.16	2.21
1994	13.10	2.21	0.60	2.18	2.24
1995	10.14	2.02	0.66	2.00	2.05
1996	10.07	2.02	0.66	1.99	2.04
1997	10.77	2.19	0.61	2.16	2.22
1998	12.60	2.26	0.60	2.24	2.29
1999	12.19	2.36	0.58	2.33	2.38
2000	12.74	2.32	0.59	2.30	2.35
2001	14.28	2.49	0.56	2.46	2.51
2002	14.89	2.49	0.57	2.47	2.52
2003	12.14	2.38	0.59	2.35	2.41
2004	9.78	2.25	0.58	2.23	2.28
2005	9.50	2.20	0.74	2.17	2.23
2006	10.66	2.29	0.72	2.26	2.32
2007	9.32	2.17	0.77	2.13	2.20
2008	11.03	2.29	0.72	2.25	2.32
2009	9.69	2.18	0.75	2.15	2.22
2010	10.97	2.28	0.73	2.25	2.32
2011	10.69	2.23	0.74	2.20	2.27
2012	9.06	2.09	0.77	2.06	2.12
Tweedie GLM					
1986	22.05	2.32	1.00	2.28	2.37
1987	17.85	2.11	0.70	2.08	2.14
1988	20.29	2.12	0.68	2.10	2.15
1989	24.08	2.28	0.59	2.25	2.30
1990	24.17	2.31	0.58	2.29	2.34
1991	21.37	2.29	0.58	2.26	2.32
1992	19.05	2.26	0.59	2.24	2.29
1993	14.89	2.18	0.62	2.15	2.21
1994	13.10	2.20	0.60	2.18	2.23
1995	10.14	2.02	0.66	1.99	2.04
1996	10.07	2.01	0.66	1.98	2.04
1997	10.77	2.18	0.62	2.16	2.21
1998	12.60	2.26	0.61	2.23	2.28
1999	12.19	2.35	0.58	2.32	2.38
2000	12.74	2.32	0.59	2.29	2.34
2001	14.28	2.48	0.57	2.45	2.51
2002	14.89	2.48	0.57	2.46	2.51
2003	12.14	2.37	0.60	2.34	2.40
2004	9.78	2.25	0.58	2.22	2.27
2005	9.50	2.19	0.75	2.16	2.23
2006	10.66	2.28	0.73	2.25	2.31
2007	9.32	2.16	0.77	2.13	2.19
2008	11.03	2.28	0.72	2.25	2.31
2009	9.69	2.18	0.76	2.14	2.21
2010	10.97	2.28	0.74	2.25	2.31
2011	10.69	2.23	0.74	2.20	2.26
2012	9.06	2.08	0.78	2.05	2.11

Table 7. Nominal and standardized CPUE estimates (numbers/hooks) for the Negative binomial GLM model with sea surface temperature and without. Least squared mean point estimated, CV% and 95% CI listed.

Neg. Bin GLM + SST					
Year	Nominal	LS mean	CV%	LCL	UCL
1986	22.05	11.76	2.97	11.10	12.47
1987	17.85	9.89	3.44	9.24	10.58
1988	20.29	10.04	3.23	9.42	10.70
1989	24.08	11.61	3.20	10.90	12.36
1990	24.17	12.27	3.17	11.53	13.06
1991	21.37	11.89	3.19	11.17	12.65
1992	19.05	11.86	3.20	11.14	12.63
1993	14.89	10.59	3.22	9.94	11.28
1994	13.10	10.85	3.23	10.19	11.56
1995	10.14	9.28	3.24	8.71	9.89
1996	10.07	9.20	3.25	8.63	9.80
1997	10.77	10.71	3.24	10.05	11.41
1998	12.60	11.63	3.24	10.91	12.39
1999	12.19	12.71	3.23	11.93	13.54
2000	12.74	12.76	3.23	11.98	13.60
2001	14.28	14.41	3.23	13.53	15.35
2002	14.89	15.23	3.24	14.29	16.23
2003	12.14	13.93	3.25	13.07	14.85
2004	9.78	12.17	3.30	11.41	12.99
2005	9.50	11.54	3.42	10.80	12.35
2006	10.66	12.48	3.40	11.67	13.34
2007	9.32	12.12	3.41	11.34	12.96
2008	11.03	13.13	3.42	12.27	14.04
2009	9.69	11.85	3.42	11.08	12.67
2010	10.97	12.80	3.43	11.96	13.69
2011	10.69	12.02	3.40	11.24	12.85
2012	9.06	11.24	3.39	10.52	12.02
Neg. Bin GLM					
1986	22.05	11.55	2.47	11.00	12.12
1987	17.85	9.70	2.99	9.14	10.28
1988	20.29	9.91	2.78	9.38	10.46
1989	24.08	11.42	2.75	10.82	12.05
1990	24.17	12.05	2.72	11.43	12.71
1991	21.37	11.71	2.73	11.10	12.35
1992	19.05	11.75	2.74	11.13	12.40
1993	14.89	10.45	2.77	9.90	11.03
1994	13.10	10.65	2.78	10.09	11.25
1995	10.14	9.03	2.78	8.55	9.54
1996	10.07	8.98	2.80	8.50	9.48
1997	10.77	10.46	2.79	9.91	11.05
1998	12.60	11.30	2.79	10.70	11.94
1999	12.19	12.36	2.78	11.70	13.05
2000	12.74	12.52	2.77	11.85	13.21
2001	14.28	14.17	2.78	13.42	14.97
2002	14.89	14.79	2.79	14.01	15.63
2003	12.14	13.59	2.80	12.87	14.36
2004	9.78	11.95	2.86	11.30	12.64
2005	9.50	11.29	3.00	10.64	11.97
2006	10.66	12.21	2.98	11.52	12.95
2007	9.32	11.88	2.98	11.21	12.60
2008	11.03	12.93	3.00	12.19	13.72
2009	9.69	11.64	2.99	10.98	12.34
2010	10.97	12.64	3.01	11.92	13.41
2011	10.69	11.88	2.98	11.21	12.59
2012	9.06	10.96	2.97	10.34	11.61

Table 8. Nominal and standardized CPUE estimates (numbers/hooks) for the Lognormal GLM model with sea surface temperature and without. Least squared mean point estimated, CV% and 95% CI listed.

Lognormal GLM+SST					
Year	Nominal	LS mean	CV%	LCL	UCL
1986	22.05	12.58	1.54	12.20	12.96
1987	17.85	10.52	1.36	10.25	10.81
1988	20.29	10.61	1.34	10.33	10.89
1989	24.08	11.89	1.30	11.59	12.20
1990	24.17	12.46	1.30	12.14	12.78
1991	21.37	12.09	1.30	11.78	12.40
1992	19.05	12.09	1.31	11.79	12.41
1993	14.89	10.95	1.33	10.67	11.24
1994	13.10	11.02	1.33	10.74	11.31
1995	10.14	9.37	1.36	9.12	9.62
1996	10.07	9.46	1.36	9.22	9.72
1997	10.77	10.75	1.35	10.47	11.04
1998	12.60	11.74	1.35	11.44	12.06
1999	12.19	12.49	1.35	12.16	12.82
2000	12.74	12.62	1.35	12.29	12.95
2001	14.28	14.63	1.35	14.25	15.02
2002	14.89	15.22	1.34	14.83	15.63
2003	12.14	14.01	1.37	13.64	14.39
2004	9.78	12.24	1.35	11.92	12.56
2005	9.50	11.15	1.60	10.81	11.51
2006	10.66	12.06	1.59	11.68	12.44
2007	9.32	11.60	1.62	11.24	11.97
2008	11.03	12.87	1.57	12.48	13.27
2009	9.69	11.65	1.60	11.29	12.03
2010	10.97	12.35	1.60	11.97	12.74
2011	10.69	11.86	1.59	11.50	12.24
2012	9.06	10.54	1.60	10.21	10.87
Lognormal GLM					
1986	22.05	12.47	1.54	12.10	12.85
1987	17.85	10.43	1.36	10.16	10.71
1988	20.29	10.55	1.34	10.27	10.83
1989	24.08	11.81	1.30	11.52	12.12
1990	24.17	12.37	1.30	12.06	12.69
1991	21.37	12.02	1.30	11.71	12.33
1992	19.05	12.05	1.31	11.74	12.36
1993	14.89	10.89	1.33	10.61	11.18
1994	13.10	10.94	1.33	10.66	11.23
1995	10.14	9.28	1.36	9.04	9.53
1996	10.07	9.38	1.36	9.13	9.63
1997	10.77	10.67	1.35	10.39	10.96
1998	12.60	11.62	1.35	11.31	11.93
1999	12.19	12.35	1.35	12.03	12.68
2000	12.74	12.54	1.35	12.21	12.88
2001	14.28	14.53	1.35	14.15	14.92
2002	14.89	15.07	1.34	14.68	15.47
2003	12.14	13.88	1.37	13.51	14.26
2004	9.78	12.15	1.35	11.83	12.48
2005	9.50	11.07	1.60	10.73	11.42
2006	10.66	11.94	1.60	11.57	12.31
2007	9.32	11.50	1.62	11.14	11.87
2008	11.03	12.79	1.57	12.40	13.19
2009	9.69	11.56	1.60	11.20	11.93
2010	10.97	12.31	1.60	11.93	12.70
2011	10.69	11.80	1.59	11.44	12.17
2012	9.06	10.42	1.60	10.10	10.75

Table 9. Nominal and standardized CPUE estimates (numbers/hooks) for the Delta GLM model with sea surface temperature and without. Least squared mean point estimated, CV% and 95% CI listed.

Delta GLM + SST					
Year	Nominal	LS mean	CV%	LCI	UCI
1986	22.05	12.39	2.74	11.72	13.05
1987	17.85	10.42	3.09	9.79	11.05
1988	20.29	10.44	3.09	9.81	11.07
1989	24.08	12.13	3.03	11.41	12.85
1990	24.17	12.72	3.00	11.97	13.47
1991	21.37	12.38	3.02	11.64	13.11
1992	19.05	12.15	3.04	11.43	12.88
1993	14.89	11.04	3.10	10.37	11.71
1994	13.10	11.31	3.07	10.63	11.99
1995	10.14	9.54	3.18	8.95	10.14
1996	10.07	9.48	3.18	8.89	10.07
1997	10.77	11.13	3.10	10.46	11.81
1998	12.60	12.13	3.04	11.41	12.85
1999	12.19	13.24	2.97	12.47	14.01
2000	12.74	13.05	2.99	12.29	13.82
2001	14.28	14.91	2.84	14.08	15.74
2002	14.89	15.45	2.80	14.60	16.29
2003	12.14	14.11	2.92	13.31	14.92
2004	9.78	12.33	3.06	11.59	13.07
2005	9.50	11.72	3.18	10.99	12.45
2006	10.66	12.62	3.12	11.85	13.40
2007	9.32	11.82	3.21	11.08	12.57
2008	11.03	12.95	3.10	12.16	13.74
2009	9.69	11.78	3.20	11.04	12.52
2010	10.97	12.87	3.11	12.09	13.66
2011	10.69	12.08	3.16	11.33	12.83
2012	9.06	10.79	3.29	10.10	11.49
Delta GLM					
1986	22.05	11.87	1.53	11.51	12.23
1987	17.85	9.75	1.38	9.48	10.01
1988	20.29	9.95	1.37	9.68	10.22
1989	24.08	11.24	1.32	10.95	11.53
1990	24.17	11.78	1.32	11.48	12.09
1991	21.37	11.40	1.33	11.11	11.70
1992	19.05	11.38	1.34	11.08	11.68
1993	14.89	10.22	1.35	9.95	10.49
1994	13.10	10.40	1.35	10.12	10.68
1995	10.14	8.61	1.40	8.37	8.85
1996	10.07	8.69	1.40	8.45	8.93
1997	10.77	10.10	1.38	9.82	10.37
1998	12.60	11.05	1.37	10.76	11.35
1999	12.19	11.88	1.37	11.56	12.19
2000	12.74	12.04	1.38	11.72	12.37
2001	14.28	14.14	1.35	13.76	14.51
2002	14.89	14.78	1.35	14.39	15.17
2003	12.14	13.58	1.39	13.21	13.94
2004	9.78	11.87	1.38	11.55	12.19
2005	9.50	10.84	1.64	10.50	11.19
2006	10.66	11.62	1.63	11.25	11.99
2007	9.32	11.25	1.69	10.87	11.62
2008	11.03	12.56	1.61	12.17	12.96
2009	9.69	11.24	1.65	10.87	11.60
2010	10.97	12.17	1.65	11.77	12.56
2011	10.69	11.55	1.63	11.19	11.92
2012	9.06	10.08	1.68	9.75	10.41

Table 10. Root mean square error (RMSE) for model fit to the true population abundance and corresponding mean coefficient of variation (CV %) for each model.

Model	RSME	CV%
Tweedie GLM + SST	0.100	0.662
Tweedie GLM + SST	0.101	0.665
Neg. Bin. GLM + SST	0.047	3.281
Neg. Bin.	0.047	2.837
Lognormal	0.071	1.422
Lognormal	0.071	1.423
GLM Delta	0.065	3.061
GLM Delta	0.061	1.454

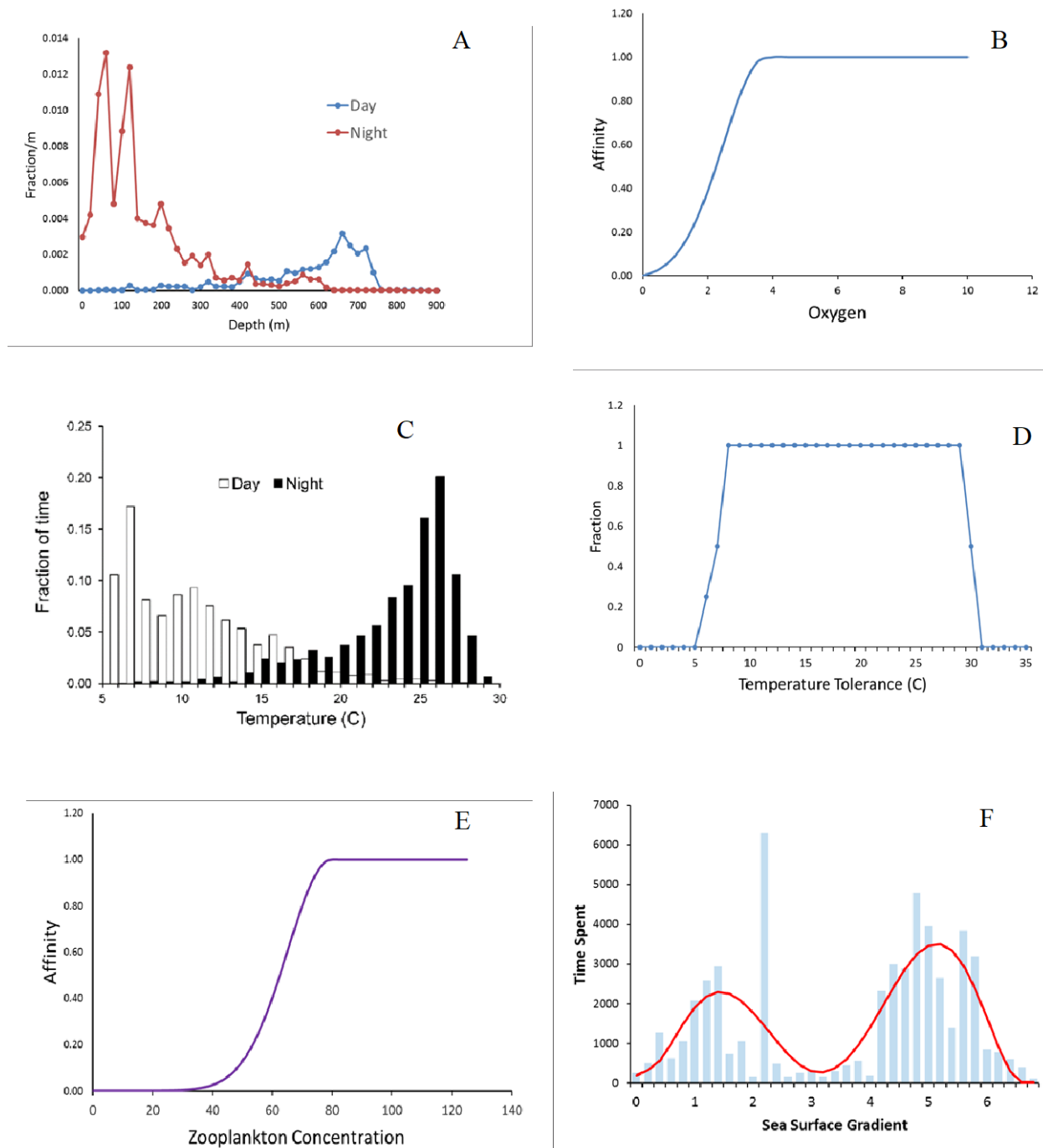


Figure 1. Affinity curves for the environmental parameters used in the swordfish SDM.

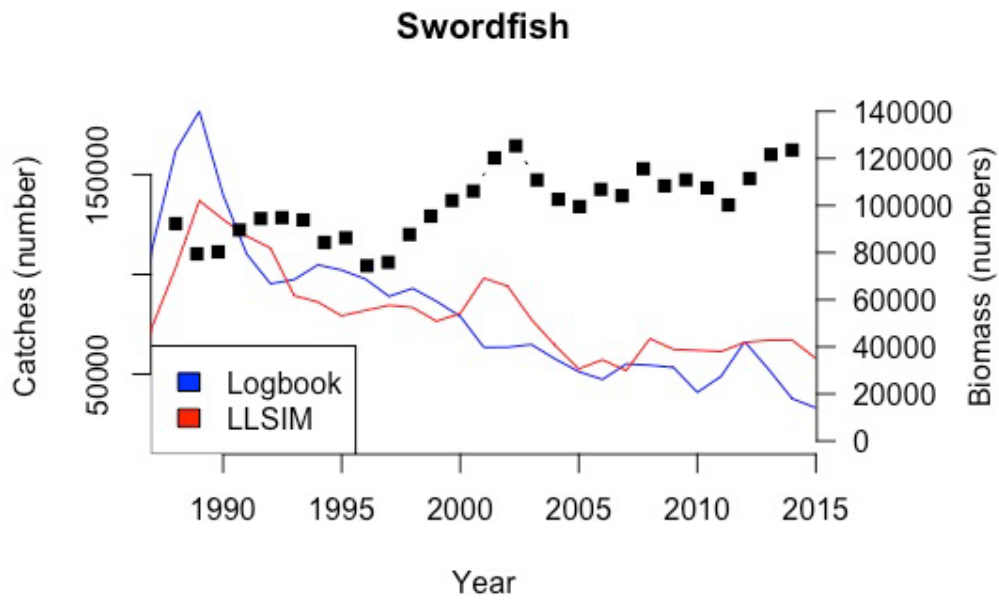


Figure 2. Catch in number of SWO from US PLL logbooks (blue) and the longline simulator (LLSIM, red). The underlying population biomass in numbers of SWO entered into LLSIM in black.

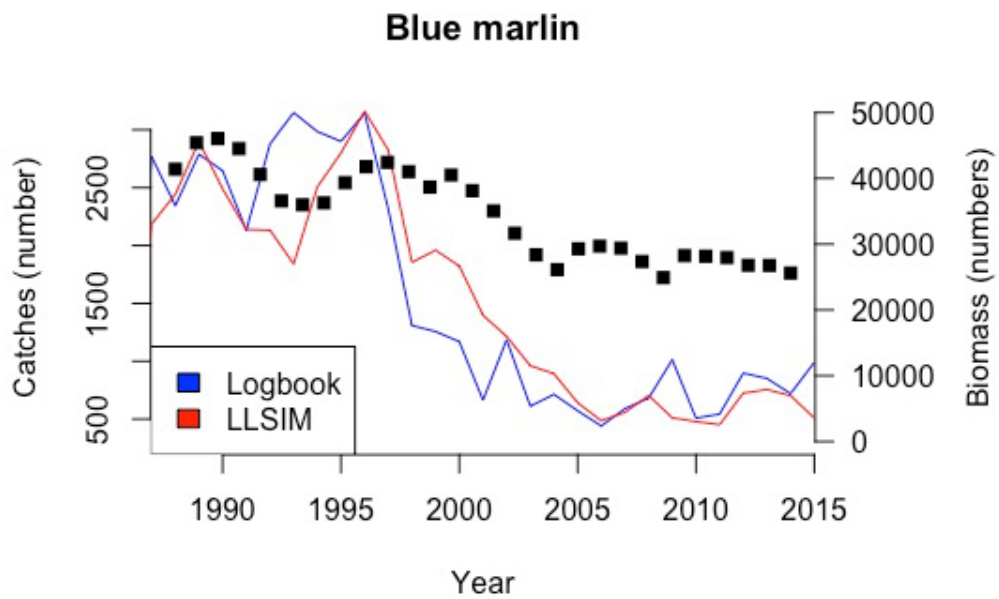


Figure 3. Catch in number of BUM from US PLL logbooks (blue) and the longline simulator (LLSIM, red). The underlying population biomass in numbers of SWO entered into LLSIM in black.

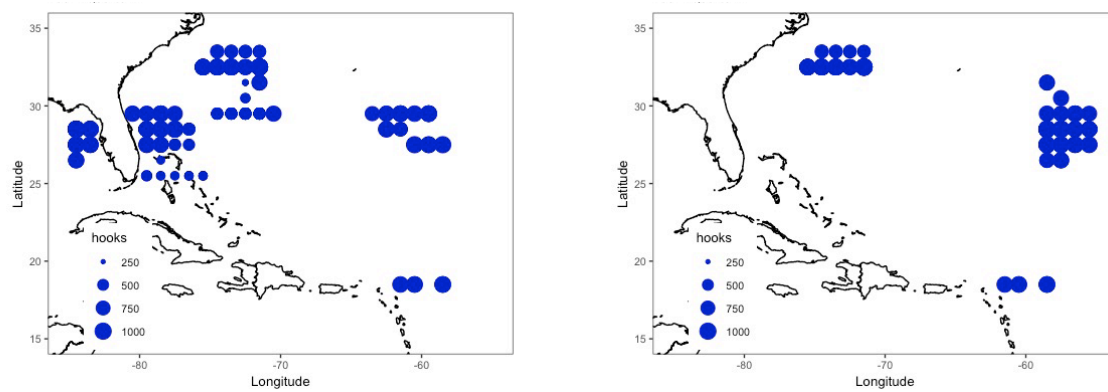


Figure 4. Effort locations for “status quo” dataset (left panel) and for closed area (right panel) in the fourth quarter of 2007.

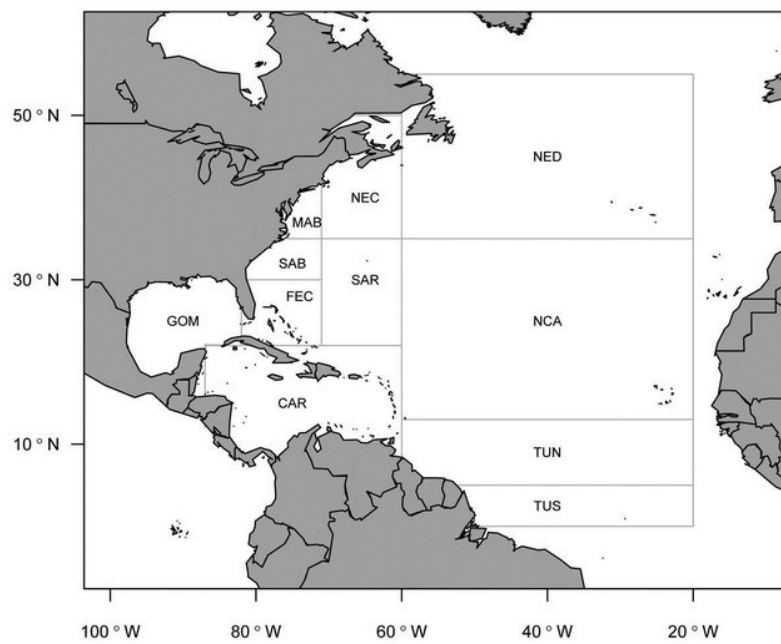


Figure 5. Geographical areas used in the CPUE standardization. Area for closed area dataset highlighted in blue.

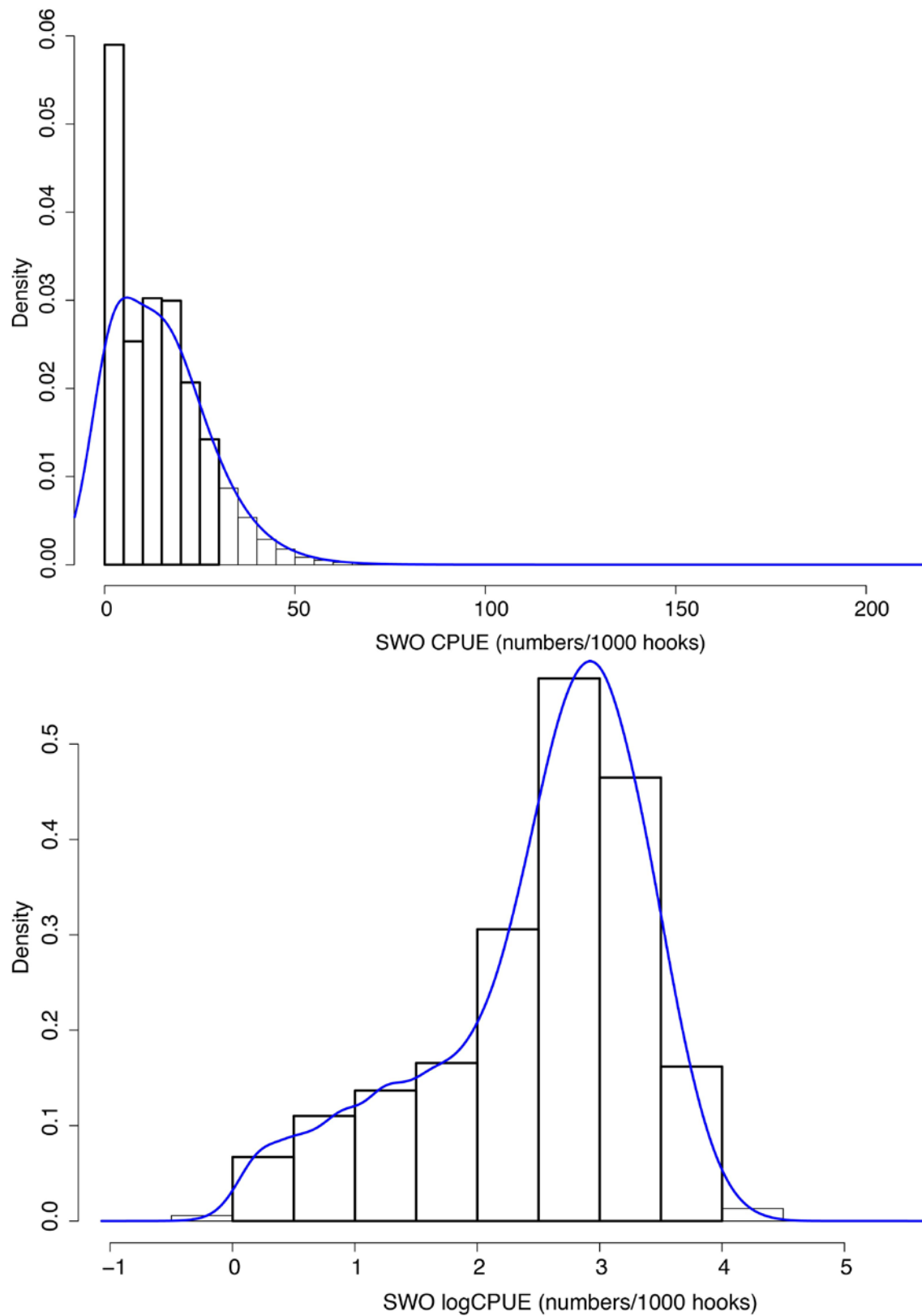


Figure 6. Distribution of nominal SWO catches from “status quo” LLSIM effort non-transformed (top panel) and log-transformed (lower panel).

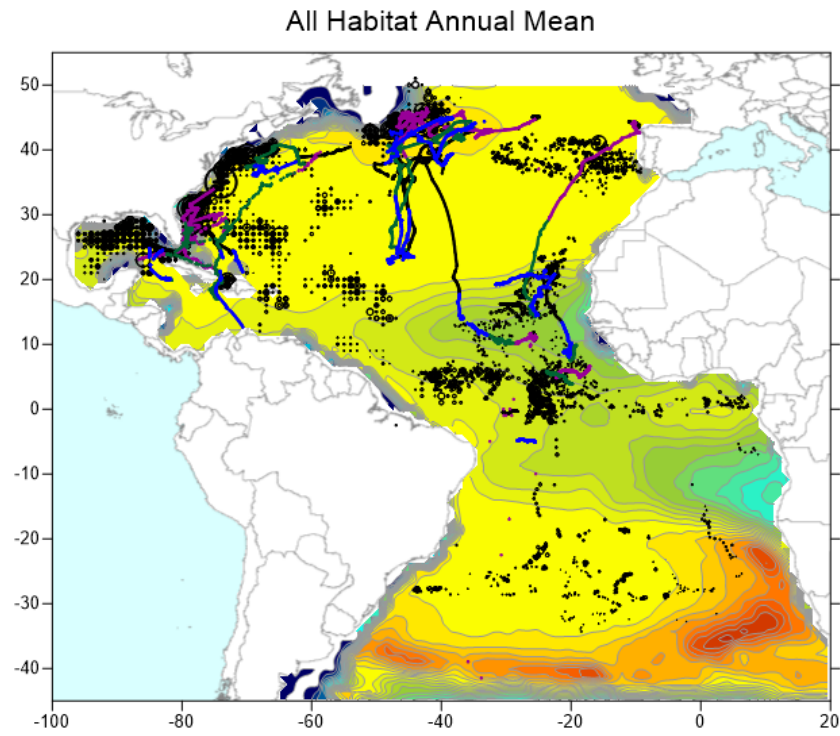


Figure 7. Distributions of average SDM predictions using temperature, oxygen, zooplankton biomass and sea surface height gradient with various CPC's observer data and PSAT tracks overlaid.

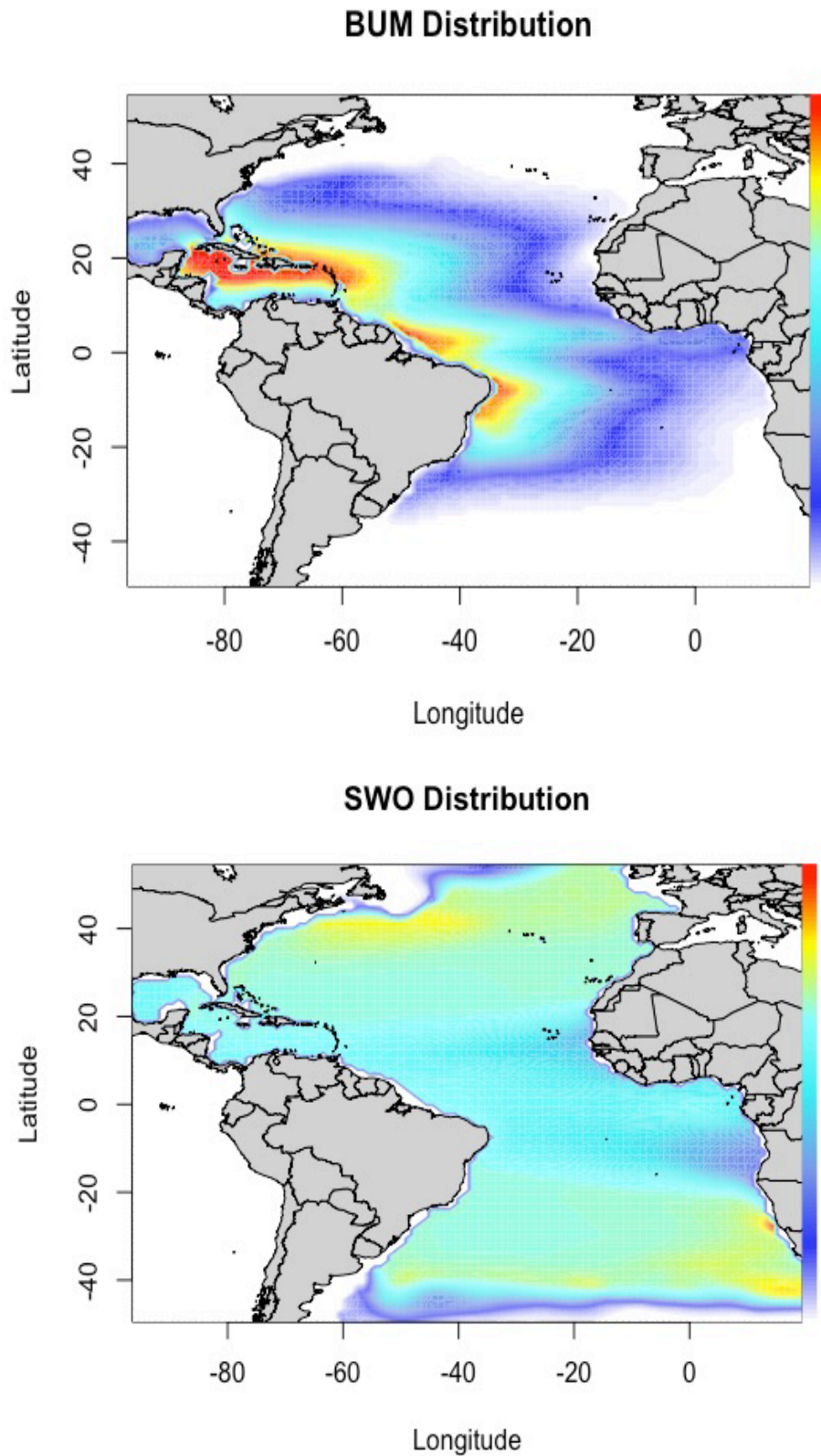


Figure 8. Distribution of blue marlin from previously published SDM (top panel, Goodyear, 2016) and swordfish distribution from this study (bottom panel).

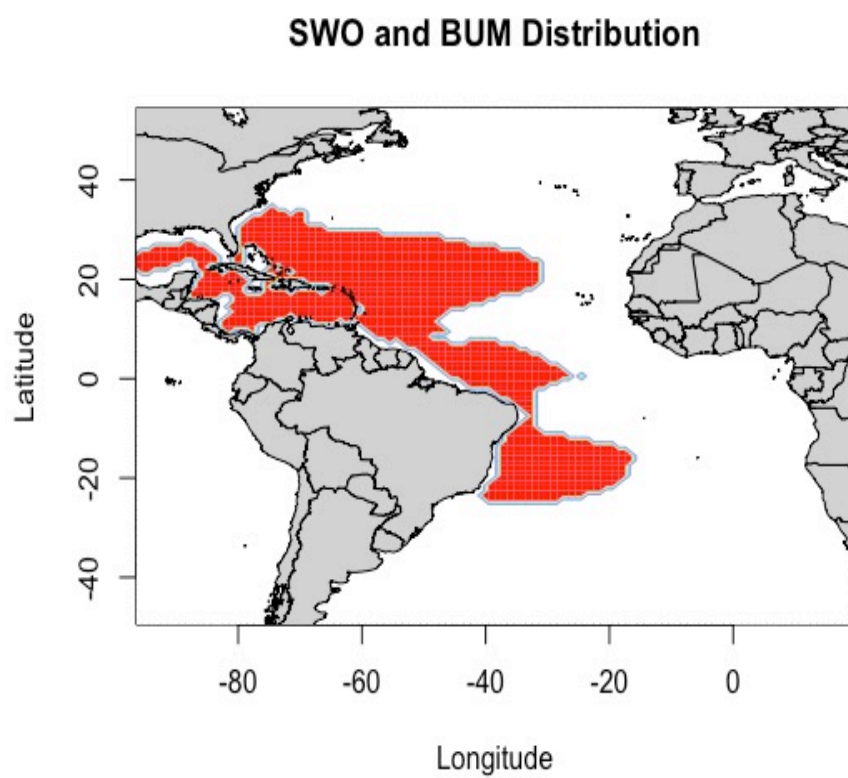


Figure 9. Areas of greatest overlap between swordfish and blue marlin averaged across all years and all months.

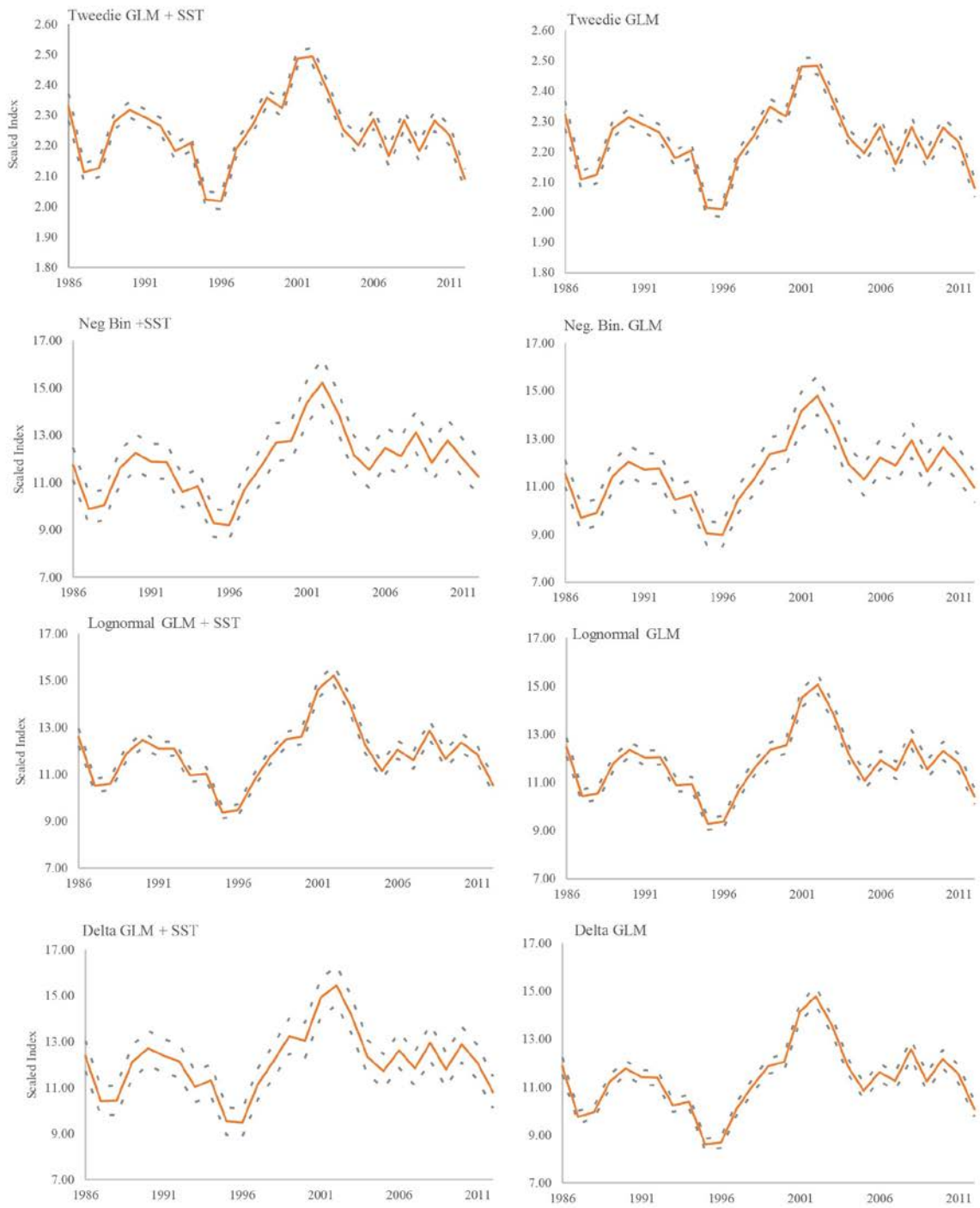


Figure 10. Standardized CPUE indices and corresponding lower and upper CI (dashed lines).

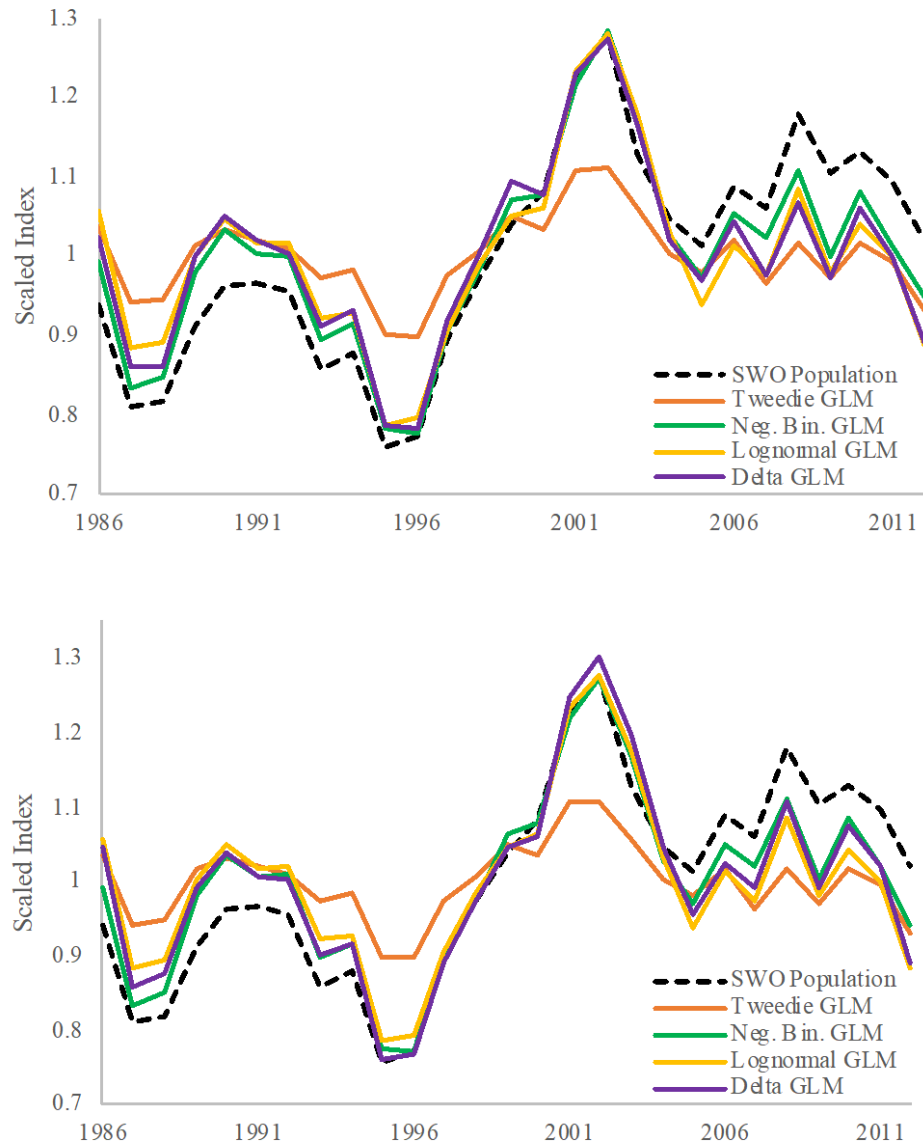


Figure 11. Standardized indices containing SST (top panel) and SST omitted (bottom panel) compared to LLSIM SWO population abundance trend.

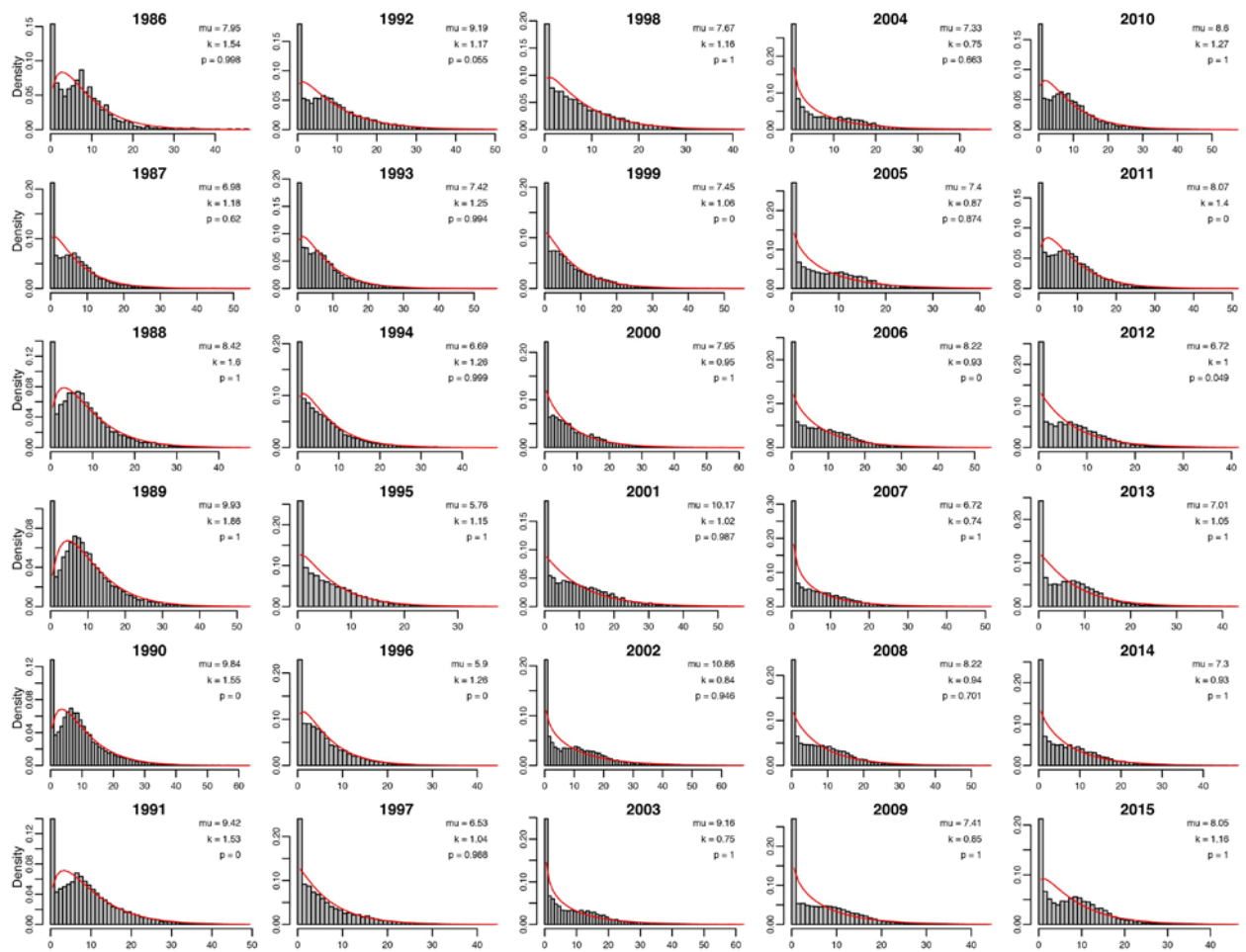


Figure 12. Yearly distribution of swordfish catches from LLSIM “status quo” effort run (in numbers) and negative binomial model fits (red line).

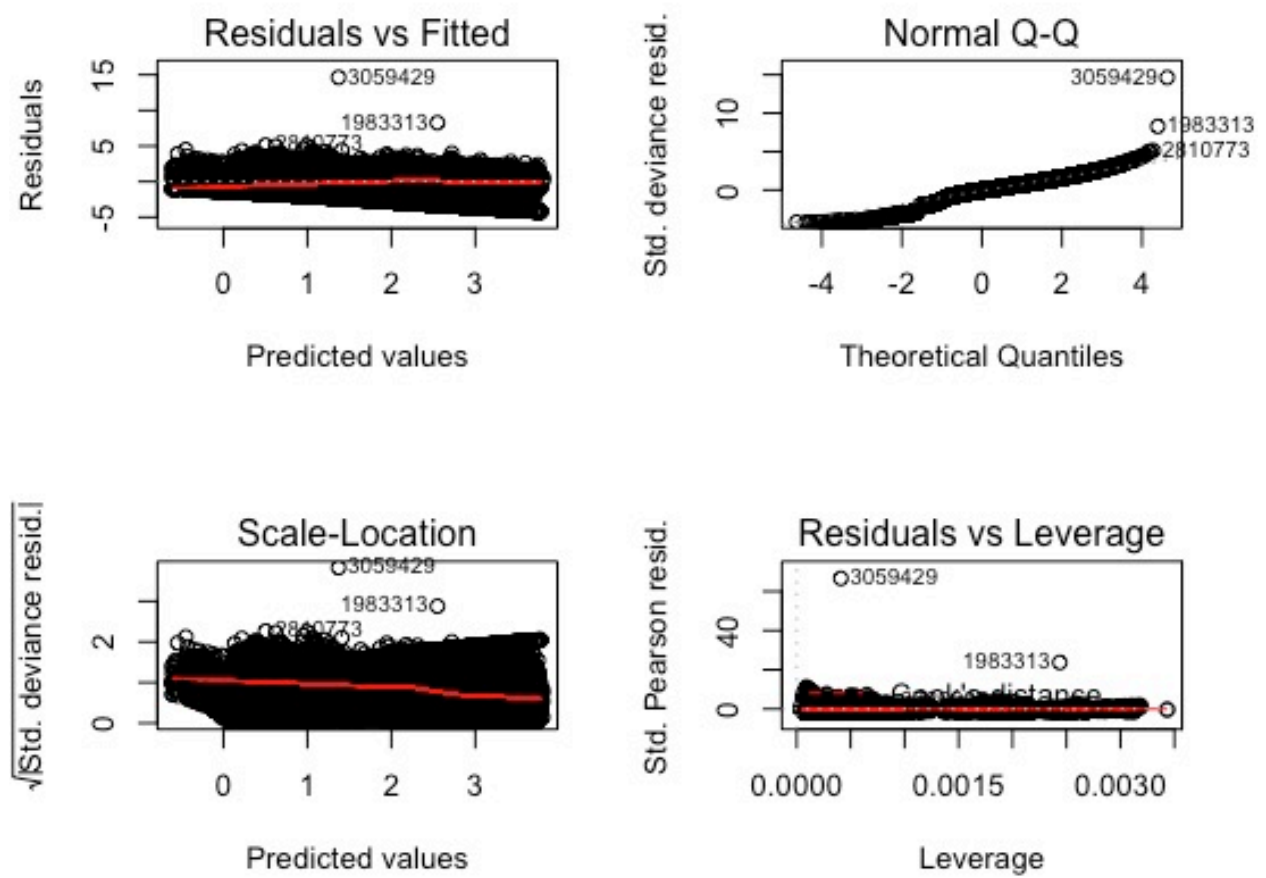


Figure 13. Diagnostic plots of negative binomial GLM.

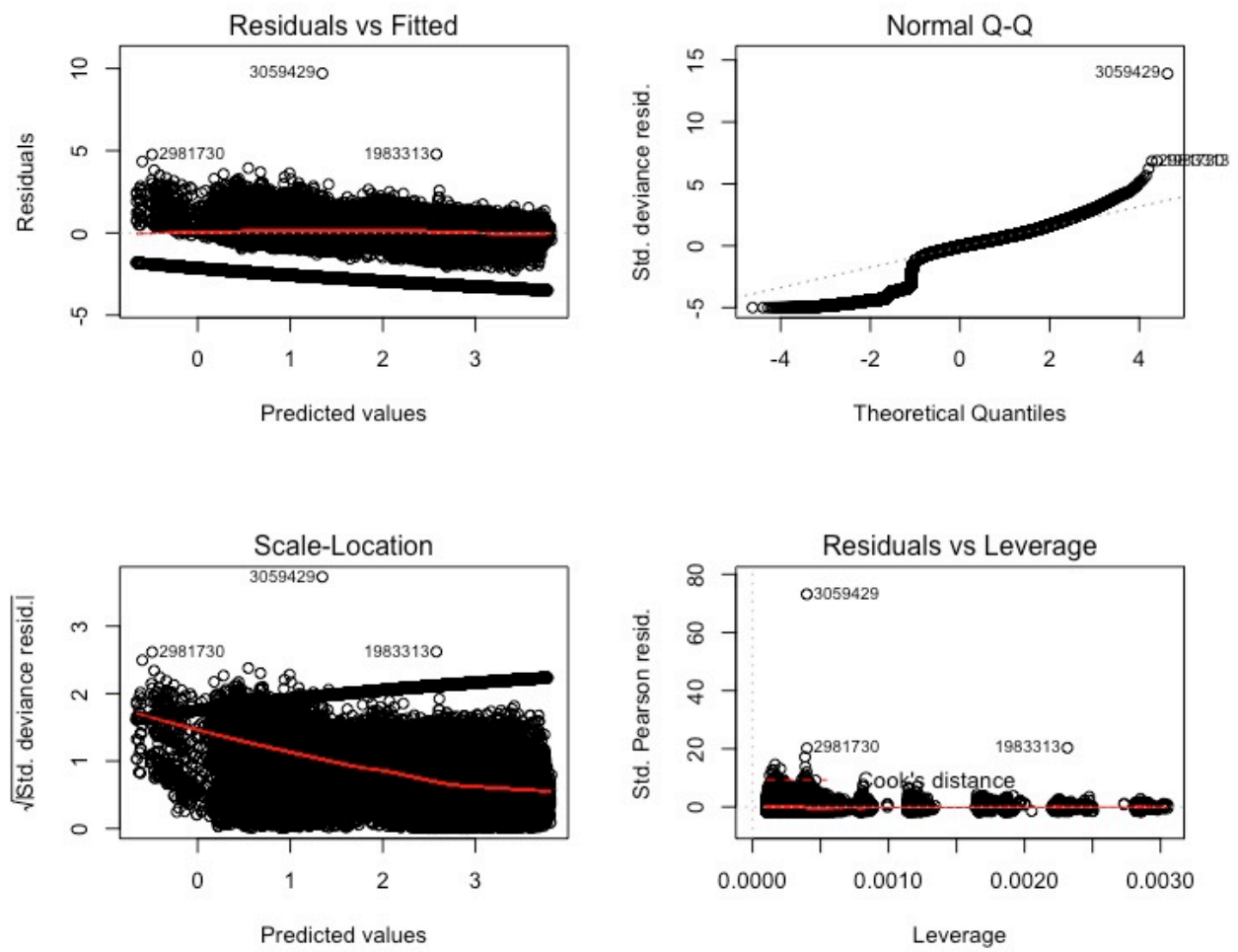


Figure 14. Diagnostic plots of the Tweedie GLM.

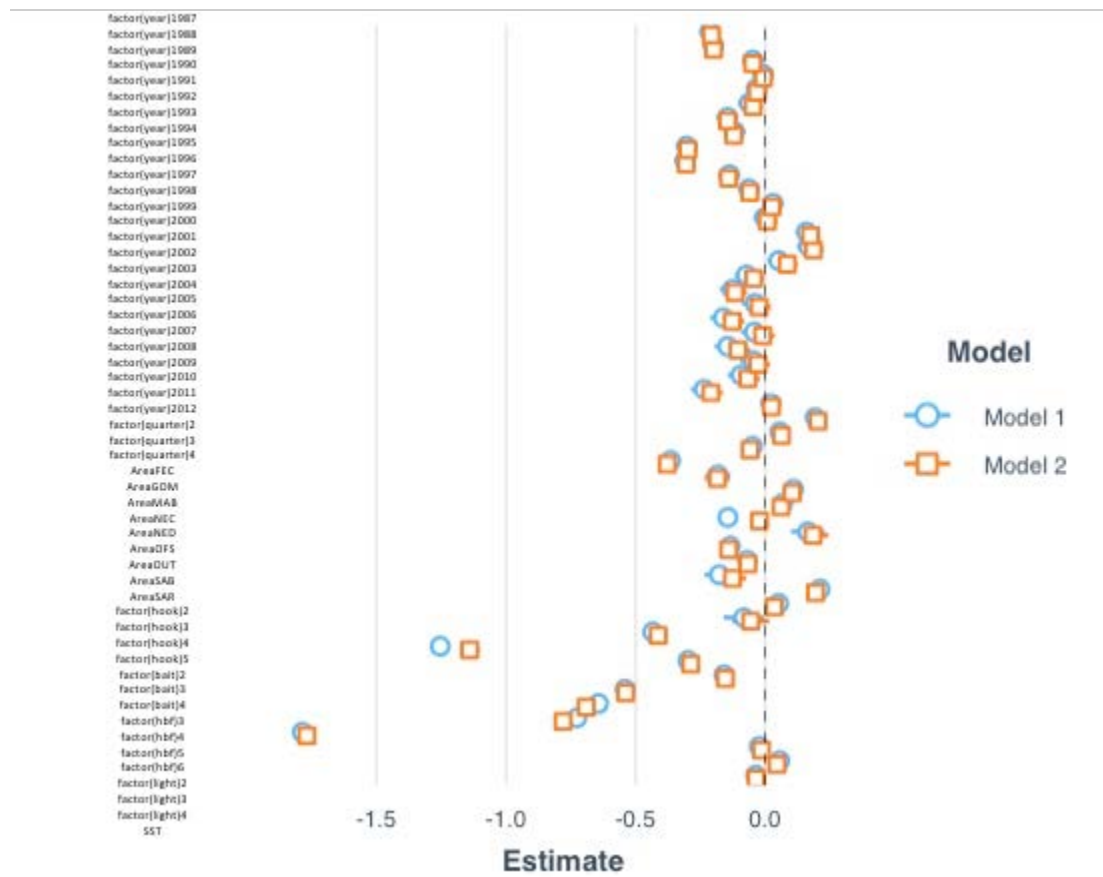


Figure 15. Estimates and uncertainties of variables in the Tweedie GLM (Model 1) and negative binomial GLM (Model 2).